

GAURAV GANDHI

Senior Data Scientist — Applied AI · GenAI & LLM Systems · Agents, RAG & Evaluation · 5 Years

Bengaluru, India | gauravgandhi429@gmail.com | +91-8789321250 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

I build AI systems and see them through — from first experiment to production. Five years across enterprise GenAI and ML: I currently lead and mentor a 5-person data-science team in Uber's AI org (via Indium), shipping document intelligence, multi-agent RAG copilots, and LLM fine-tuning pipelines — including a document-understanding transformer pretrained on 50M+ documents that displaced manual review for \$10M+ in annual savings. Outside work I run nine live AI products and publish research on agent tool-use quality. I care most about honest evaluation: every number I ship is measured, and I say so when a simple baseline wins.

PROFESSIONAL EXPERIENCE

Indium Software Client: Uber Technologies — Uber AI

Jun 2024 – Present

Senior Data Scientist

May 2025 – Present · Bengaluru, India

- Lead and mentor a 5-person data-science team across GenAI document-intelligence and conversational-AI workstreams — owning the modeling roadmap, evaluation strategy, and technical review.
- **LLM fine-tuning for document intelligence:** raised field-level extraction accuracy to 97%+ with a parameter-efficient fine-tuning framework (Llama 3.1, GPT-Neo; LoRA/QLoRA, layer freezing, instruction-tuning data design) and an automated field-accuracy eval harness; mentored multiple teams on prompt and evaluator design.
- **Vision document-quality gate (ViT):** removed the manual-review bottleneck in e-commerce document verification — a Vision Transformer scores each upload against 9 rejection reasons across 55 document types, auto-approving only below a calibrated per-(type, reason) human-recall threshold.
- **Multi-agent GenAI copilot (hybrid RAG + NL2SQL):** cut analytics turnaround from hours to seconds for non-technical teams — dense + sparse retrieval, NL2SQL over a schema knowledge graph, LangChain tool orchestration; live across business units.

Data Scientist

Jun 2024 – May 2025 · Bengaluru, India

- **Document-understanding transformer at production scale:** owned the multi-task training objective and extraction head for an encoder-decoder transformer pretrained on 50M+ documents across 144 A100 GPUs (PyTorch, Ray Train, DeepSpeed ZeRO-3) — shipped at 95%+ field accuracy, displacing manual review for \$10M+ in annual cost savings.
- **Session-aware recommendations (BERT):** drove a sustained 27% lift in user engagement by replacing static collaborative filtering with a session-aware sequence model capturing evolving intent — validated via A/B tests across releases.

FedEx Express Consumer Logistics & Supply Chain

Aug 2022 – Jun 2024

Decision Scientist

Bengaluru, India (Remote)

- **Probabilistic anomaly detection for account retention:** recovered lost enterprise shipping volume, surfaced via average-daily-volume decline, with an ensemble of Bayesian change-point detection, probabilistic clustering, and profitability scoring — shaping regional supply strategy.
- **Demand forecasting (SARIMA + Bayesian tuning):** reduced operational inefficiencies ~20% by upgrading ARIMA pipelines to SARIMA with Box-Jenkins tuning, Fourier seasonality, and automated anomaly filtering.

Tata Consultancy Services (TCS)

Jul 2021 – Jul 2022

Data Engineer

Bengaluru, India (Remote)

- **High-volume data pipelines (GCP):** designed end-to-end ETL pipelines and analytics dashboards for a high-volume booking/transaction client — campaign targeting, segmentation, trend monitoring.

RESEARCH

“Tool-Description Quality Is Not One Axis: A Regime Analysis of Where It Helps and Where It Backfires” — first-author preprint (arXiv ID pending). Built AgentGauge, the statistical harness behind it — its own predictive-validity study falsified the v1 quality score; the rebuilt causal A/B harness measures a BLOCKING description defect at -13.3 to -28.9pp agent task success (3 model families); frozen protocol, CI-enforced. github.com/gaurav-gandhi-2411/agentgauge

SELECTED INDEPENDENT PROJECTS

Nine live AI products, built and run end-to-end under my own name. Every metric below is reproduced in the linked repo; the full set with sources is on the portfolio.

- **Warmer — daily semantic word game** [Live](#) — built the full product (Flutter, sentence-transformers, Firebase), running live daily. Diagnosed Hinglish embeddings scoring near-random as a script-mismatch bug, then trained and published the LoRA fine-tune that now serves production (hinglish-relatedness-sbert on Hugging Face): Spearman $-0.003 \rightarrow 0.813$, held-out $0.435 \rightarrow 0.704$.
- **Style Maitri — AI stylist for Indian occasions** [Live](#) [GitHub](#) — LangGraph agent over hybrid retrieval (FAISS + BM25, reciprocal-rank fusion) across $\sim 52K$ items from 8 store catalogues, with sentence-level grounding that scrubs hallucinated price/size claims. Intent-parsing accuracy 93.8% ($n=211$).
- **TriagelQ — ML issue-triage service** [Live](#) [GitHub](#) — four-stage triage (TF-IDF classifier, BGE+FAISS retrieval, LightGBM estimate, LLM synthesis); component classifier top-3 accuracy 82.5% (k8s) / 90.4% (vscode). Built the disjointness guard that caught 3 months of silent train/eval contamination, then audited and published the LLM stage's fabrication rate (1.9–9.1%) rather than hiding it.
- **DealHunter — multi-agent flight search** [Live](#) [GitHub](#) — natural-language trip requests to Pareto-ranked itineraries via SSE streaming — planner/optimizer agents with an LLM-as-judge eval harness; 597 tests, $\geq 87.65\%$ coverage.
- **Multimodal Fashion Recommender** [Demo](#) [GitHub](#) — two-tower model fusing CLIP image and SBERT text embeddings (in-batch InfoNCE); Recall@10 $3.06\times$ over a popularity baseline (0.0328 vs 0.0107), validated by modality ablation.
- **AetherArt — diffusion art within a consumer-GPU budget** [Live](#) [GitHub](#) — custom rank-8 Ukiyo-e SDXL LoRA composed with Hyper-SD and ControlNet, served on Cloud Run — full pipeline at 6.2GB peak VRAM against an 8GB budget.
- **Also:** ReviewIQ (multilingual review extraction — 83.8% accuracy on a CI-gated eval); tracegauge (published PyPI package scoring AI-session efficiency); Gold Rate Tracker (the forecaster that ships its naive baseline, because the baseline honestly beat the model).

KEY TECHNICAL SKILLS

- **LLMs, GenAI & Fine-Tuning:** LLM APIs (OpenAI, Anthropic, Gemini, Groq, Hugging Face), LoRA/QLoRA, PEFT, instruction tuning, RLHF/DPO, prompt engineering, guardrails & grounding
- **Evaluation & LLM Observability:** LLM-as-judge design, eval-harness & fixture design, CI-gated regression evals, contamination audits, A/B testing, MLflow, Prometheus, structured logging
- **RAG, Retrieval & Agents:** hybrid retrieval (dense + BM25/RRF), FAISS, pgvector, Pinecone, NL2SQL, knowledge graphs, multi-agent systems, LangChain, LangGraph, MCP
- **ML & Deep Learning:** transformers (BERT, GPT, Llama, CLIP, ViT), diffusion (SDXL), two-tower/contrastive (InfoNCE), recommenders, sequence models, Bayesian inference, time-series
- **Modeling Stack & Engineering:** PyTorch, Hugging Face (transformers, diffusers, PEFT, sentence-transformers), scikit-learn, LightGBM, Python (advanced), SQL, PySpark, FastAPI, unit & integration testing
- **Deployment, MLOps & Distributed Training:** Docker, GCP (Cloud Run, Vertex AI), Azure, GitHub Actions, Vercel, Firebase; Ray Train, DeepSpeed ZeRO-3

EDUCATION

Post Graduate Diploma in Data Science — Great Lakes Institute of Management Feb 2022 – Nov 2022
GPA 8.45/10.0 · Chennai, India (Remote)

B.Tech, Electronics & Communication Engineering — Manipal University Jaipur Jul 2017 – May 2021
GPA 8.05/10.0 · Jaipur, India

CERTIFICATIONS

NLP Specialization · Deep Learning Specialization · LLMops Specialization (DeepLearning.AI) · Google Cloud Professional Data Engineer